Routed Detector Ensembles with Validation Noise-Bank Verification for Noise Event Detection in Vaani Speech

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***Abstract*—Track 1 of the IndoML 2026 Datathon asks for the onset and offset of every non-speech noise event in Vaani speech recordings, scored by Event F1 plus Segment Dice. We train 24 frame-level detectors on the Vaani Noise Event Dataset (Whisper, WavLM and BEATs encoders with recurrent heads, and log-mel CRNNs) with a multi-tier loss that combines verified, unverified and clip-level labels, plus synthetic clips whose boundaries are exact. A random-forest router sends clips that look organiser-synthesised to a synthetic ensemble branch and the rest to a natural branch with boundary post-processing. The synthetic test clips reuse noise recordings from a finite pool that the official validation release exposes (mixture minus clean reference); we locate these recordings in the test audio by cross-correlation and let energy-verified matches set exact event boundaries, a procedure the organisers approved in writing. Our Track 2 enhancer serves as a noise meter that removes speech-only false alarms. Team CodeAMU ranked 1st on the final (Phase 2) Track 1 leaderboard with Combined 1.6173 (Event F1 0.7895, Segment Dice 0.8278; synthetic clips F1 0.998, natural clips F1 0.520).**

***Index Terms*—sound event detection, noise events, Whisper, ensembles, cross-correlation, Vaani**

I. Introduction

Each test clip (16 kHz; 5,517 clips, 10.8 h) must be annotated with the intervals of its non-speech noise events. Event F1 matches predictions one-to-one to references when both onset and offset lie within $\max(0.2\,d, 50\,\mathrm{ms})$ of the reference (d: reference duration) [3]; Segment Dice is computed per clip on a 10 ms grid and averaged over clips; Combined = F1 + Dice. The test set mixes natural recordings with organiser-synthesised clips (clean speech plus pasted noise), which behave so differently that our system treats them separately.

Final standing: **1st place** on the Phase 2 final leaderboard, Combined **1.6173** (Event F1 0.7895, Segment Dice 0.8278; natural clips F1 0.5197 / Dice 0.7256; synthetic clips F1 0.9981 / Dice 0.9966). Training and inference code, configurations and all checkpoints are released in our public repository `CodeAMU_Track1`.

II. Training Data and Preprocessing

A. Vaani Noise Event Dataset

We train on the Vaani Noise Event Dataset [1] (Hugging Face ARTPARK-IISc/Vaani-Noise-Event-Dataset), derived from Project Vaani [2]: 90,635 clips (154.6 h) of multilingual Indian speech in three annotation tiers, *gold* (verified timestamps, 11,111 clips), *silver* (unverified timestamps, 61,640) and *bronze* (clip-level labels only, 17,884). Category strings are mapped to seven classes (animal, vehicle/traffic, baby/child, singing/music, phone/signal/alarm, appliance/machine, human non-speech). Audio is decoded from the parquet shards to 16 kHz mono WAV; training crops are at most 10 s and frame targets lie on a 20 ms grid (10 ms for two *hr* models). Two gold hold-out sets (1,111 and 3,999 clips) serve for model selection.

B. Official validation release

The 2,501 validation clips (1,806 natural, 695 synthetic; each synthetic clip ships its clean reference) are the only additional data. We split them into halves A/B (1,259/1,242 clips) and use them (i) as extra gold data: fold models are trained on gold + one half and evaluated on the other, and three *full-gold* models use all 13,612 gold and validation clips; (ii) to train the router and boundary regressor and to tune post-processing out-of-fold; (iii) as the source of the noise bank (Sec. III-C). No other external data is used.

C. Synthetic augmentation

To obtain clips whose boundaries are exact by construction we synthesise 11,762 clips (21.4 h, 21,095 events). From the 7,104 gold clips of our training split we harvest a noise bank (the 12,585 annotated event spans, 4.0 h) and a clean bank (5,884 event-free gaps, 6.5 h). Each clip pastes noise segments onto a clean bed at SNR $\sim \mathcal{U}(-22, -2)$ dB with 15 ms cosine fades; 35% of the inter-event gaps are drawn from 50–300 ms (the rest from 0.4–2.5 s), because merging close events was our most frequent error. The *syn* base detectors train on all 90,635 clips plus these 11,762.

III. Proposed Method

A. Detectors

Every detector maps a waveform to 20 ms frame logits through five heads on a shared sequence representation: noise presence; onset and offset (targets are Gaussian bumps, $\sigma = 40$ ms, at gold boundaries); a 7-way frame category; and a clip-level weak head with linear-softmax attention pooling [10], through which bronze clips train. Encoders (Table I): Whisper [4] encoders with positional embeddings sliced to the clip length and WavLM Base+ [5], both with a learnable weighted

TABLE I

THE 24 DETECTORS. SYN: BASE MODEL ON ALL TIERS + SYNTHETIC CLIPS; SPLIT: BASE MODEL ON ALL TIERS MINUS A GOLD HOLD-OUT SET; FT: GOLD-ONLY FINE-TUNE; A/B: GOLD + ONE VALIDATION HALF (SCORED ON THE OTHER); FULL: ALL GOLD + VALIDATION; 10MS: 10 MS FRAMES.

Encoder	Params	#	Variants
Whisper large-v3	640.6M	7	split, split+FT, syn, syn+FT, A, B, full
Whisper large-v2	640.4M	2	split+FT, full
Whisper medium	310.4M	5	syn, A, B, syn-10ms, syn-10ms+FT
Whisper small	90.9M	3	syn, A, B
WavLM Base+	97.2M	1	split+FT
BEATs + Whisper-m.	405.1M	2	split+FT, full
CRNN (log-mel)	0.8/4.4M	4	split+FT, syn ($\times$2), syn+FT

sum of hidden layers [8] and a 2-layer BiGRU (256 units); a hybrid in which frozen BEATs [6] features, frequency-pooled and upsampled $\times 8$, are concatenated with a Whisper-medium stream; and log-mel CRNNs trained from scratch [9] (64 or 128 mels, 3–4 convolution blocks, BiGRU). The loss is

$$\mathcal{L} = \mathcal{L}_{\text{gold}} + \lambda_s \mathcal{L}_{\text{silver}} + 0.3\,\mathcal{L}_{\text{cat}} + 0.5\,\mathcal{L}_{\text{weak}} + \mathcal{L}_{\text{on}} + \mathcal{L}_{\text{off}}, \quad (1)$$

all binary cross-entropies: presence terms use the gold or silver timestamps ($\lambda_s = 0.25$, 0 in gold-only fine-tuning), boundary terms use gold clips only. Training uses AdamW [11] (weight decay 0.01), head learning rate $5{\cdot}10^{-4}$–10^{-3}, encoder learning rate $6{\cdot}10^{-6}$–$3{\cdot}10^{-5}$ with the encoder frozen for the first epoch, bf16, 4–34 epochs, and keeps the best checkpoint by validation Combined. Each base model is then fine-tuned on gold data only (learning rate $2{\cdot}10^{-4}$, 2–12 epochs), on gold + validation half A or B, or on all gold + validation. The BEATs base run uses mean-teacher consistency [12] (EMA 0.999), the WavLM base run stochastic weight averaging [13].

B. Routing and branch ensembles

Organiser-synthesised clips leave traces between events: a quiet, spectrally flat background, little high-band energy in quiet frames, a steady noise floor and abrupt energy jumps at splice points. A random forest [14] on 24 such background statistics, trained on the `syntheticData` flags of the validation release, routes 2,080 of the 5,517 test clips to the synthetic branch.

Natural branch. Presence and boundary tracks are weighted means of 16 detectors (weights 0–1.4, chosen out-of-fold separately for presence and boundaries), blended with the three full-gold models (1/19 each) and two CRNNs (0.20, 0.15). Decoding uses hysteresis thresholds 0.57/0.48, a 3-frame median filter and a 0.1 s minimum duration; events are split at onset/offset peak pairs, edges are snapped to boundary peaks by duration, long events are carved, and a gradient-boosting boundary regressor [15] (Huber loss, trained on the validation folds) trims offsets by 5–200 ms. Post-processing removes duplicates (IoU-based suppression), drops low-confidence false alarms by rules on mean and peak presence, cuts offsets at the steepest presence descent and shifts collars by +4/+8.5 ms. *Synthetic branch.* Unweighted mean of 10 detectors with its own thresholds, splitting and snapping. Every parameter was tuned on validation halves A/B and kept only if both halves improved.

C. Noise-bank verification (disclosure)

The synthetic test clips paste noise from a finite pool, and the validation release exposes that pool: *noise* = *mixture* − *clean* at the labelled timestamps yields 2,213 events, i.e. 1,181 distinct recordings after de-duplication (4,118 templates when the labelled events of the natural validation clips are added). We pre-emphasise (0.97) and resample to 8 kHz, cross-correlate each test clip with every unit-norm template by FFT, normalised by the local signal energy [16], keep up to three peaks per template with correlation ≥ 0.20, and pick matches greedily by explained energy (overlap $\leq 30\%$). A match at lag t gives the span $[t, t+\ell]$; for cross-clip copies within the validation set it equals the annotation in 1,925 of 1,926 cases at correlation ≥ 0.6 (median edge error 0.3 ms). Truncated or false matches are rejected by an energy test: after subtracting the template with a fitted gain, some 10 ms frame must lose at least 1 dB, and the first and last 100 ms must lose at least 0 dB at correlation 0.45–0.6, or at least -1 dB above 0.6; at 0.30–0.45, at least 30% of the template frames must also be explained. Trusted spans snap nearby predictions, replace overlapping predictions that miss the tolerance, and add missed events, in synthetic clips and, since the pool was cut from natural recordings, in natural clips too (correlation ≥ 0.7, 94–96% exact on validation). The organisers confirmed in writing that noise waveforms derived from the validation release may be located in test audio by cross-correlation to determine Track 1 event boundaries, provided the method is disclosed, as done here.

D. Noise meter and empty clips

Seventy-five synthetic predictions overlap no bank span. We run our Track 2 TF-DPCRN enhancer [17] over the clip and measure the removed-energy share $D = 10\log_{10}(E_{\text{removed}}/E_{\text{mixture}})$ over each event: on bank spans (true noise) the median is -2 dB, on noise-free speech -64 dB; the 28 events with $D < -12$ dB are dropped. Segment Dice gives a clip without predictions a score of 0, so each of the 31 natural test clips left empty receives one event spanning the natural ensemble's activity envelope (first to last frame with $p \geq 0.20$).

E. Implementation and pretrained models

PyTorch 2.11 [18] with CUDA 12.8, `transformers` [19] and scikit-learn [20], on NVIDIA H200 GPUs (141 GB; base runs on a full GPU, fine-tunes on 71 GB MIG slices). Inference uses 10 s windows with an 8 s hop in bf16; the 24 detectors process 10.9 h of audio in about 50 min on one 71 GB MIG slice (three detectors in parallel); bank matching and post-processing run on CPU. Table II lists every pretrained checkpoint; all 24 detector checkpoints, the router and the boundary regressor are released with the code.

TABLE II
PRETRAINED MODELS (HUGGING FACE; FIRST 7 CHARACTERS OF THE REVISION).

Checkpoint	Rev.	Role
`openai/whisper-small` [4]	973afd2	encoder, 3 detectors
`openai/whisper-medium` [4]	abdf7c3	encoder, 7 detectors
`openai/whisper-large-v2` [4]	ae46427	encoder, 2 detectors
`openai/whisper-large-v3` [4]	06f233f	encoder, 7 detectors
`microsoft/wavlm-base-plus` [5]	4c66d48	encoder, 1 detector
BEATs iter3+ AS2M cpt2 [6], [7]	172e407	frozen, 2 detectors
CodeAMU Track 2 TF-DPCRN [17]	ours	noise meter

TABLE III
LEADERBOARD PROGRESSION ON THE 5,517 TEST CLIPS (PHASE 2 FINAL RANKING USES THE SAME TEST SET).

Submission	Comb.	F1	Dice	$F1_{nat}$	$F1_{syn}$
v60: routed ensembles (Phase 1 final)	1.5985	.7773	.8212	.5096	.9865
v67: + synthetic duration snapping	1.6001	.7788	.8213	.5096	.9892
v68: + bank snaps, synthetic	1.6066	.7812	.8254	.5096	.9936
v93: + natural bank merge, post-proc.	1.6134	.7879	.8255	.5193	.9946
+ energy test, noise meter	1.6158	.7897	.8262	.5193	.9980
+ bank spans in natural clips	1.6167	.7903	.8264	.5208	.9981
+ empty-clip fill (final, 1st)	**1.6173**	.7895	.8278	.5197	.9981

IV. RESULTS AND CONCLUSION

Table III traces the leaderboard. Routing is the largest single factor: our early submissions, made before routing, scored 1.15–1.17 with synthetic-subset F1 below 0.38; routed ensembles reached 1.55–1.60. Noise-bank verification then lifts synthetic F1 from 0.989 to 0.998 and synthetic Dice from 0.983 to 0.997, so the synthetic clips are essentially solved. On held-out validation data, single detectors score 1.38–1.45 Combined (Whisper large-v3 1.42–1.45, Whisper small 1.40–1.42, CRNN 1.41), and the natural branch scores 1.173 (F1 0.462, Dice 0.711) out-of-fold on the 1,806 natural clips.

Natural clips are the bottleneck. Out-of-fold, 43% of reference events are matched, 46% fail on a boundary (an edge outside the tolerance, offsets more often than onsets) and only 10.5% are missed. Environmental sounds (vehicle, animal, baby) have scattered rather than biased edges (median error 90–150 ms against 23–39 ms for human non-speech), which points to ambiguous annotation of gradual sounds. Correcting both edges of near-misses would be worth +0.52 Combined out-of-fold, whereas removing pure false alarms would be worth only +0.02. Ideas that lost out-of-fold include learned false-alarm pruning (Dice is averaged per clip, so deleting an overlapping event costs more Dice than it gains F1), extra ensemble members, enhancer-based boundary snapping, and extra events in non-empty clips.

In conclusion, specialised ensembles behind a learned router, combined with exact verification against the validation noise bank, gave the winning Track 1 system (1st place, Combined 1.6173). The synthetic part of the task is solved by matching; the remaining headroom lies in natural-clip boundaries, which call for consistent boundary annotation more than for larger models.

Reproducibility. From the cached detector outputs and noise-bank matches, the released pipeline regenerates every intermediate submission, from v67 to the final one, identically on all 5,517 clips; re-running four of the 24 detectors (one per architecture type) reproduces their cached frame probabilities bit for bit.

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